

07 — Install and use a conda environment on Windows /

07 — 在 Windows 安装和使用 conda 环境

Question and scope / 问题与范围 (Question and scope)

How do I install conda if necessary, create a separate Python environment, and use the correct interpreter in VS Code?

如果尚未安装 conda，如何安装？如何创建独立 Python 环境，并让 VS Code 使用正确的解释器？

Anaconda/Miniconda supplies conda, the environment and package manager. A conda environment holds a Python interpreter and its packages. VS Code is the editor; choosing an interpreter connects it to an environment. This Windows walkthrough uses a new environment named `handbook-demo`; it does not change an existing research environment. Commands are instructions for you to run, not a record that an installation was performed. Documentation was checked on 2026-09-20.

Anaconda/Miniconda 提供环境和包管理器 (environment/package manager) conda；conda 环境包含 Python 解释器 (interpreter) 及软件包。VS Code 是编辑器，选择解释器才能连接到相应环境。本篇面向 Windows，以新环境 `handbook-demo` 为例，不改动已有科研环境。命令是供你执行的步骤，不代表已经完成安装。官方资料核对日期：2026-09-20。

1. Check whether conda is already installed / 1. 检查是否已经安装 (Installation check)

Open **Anaconda Prompt** from the Windows Start menu and run:

从 Windows 开始菜单打开 **Anaconda Prompt**，执行：

```
conda --version
conda env list
```

If both work, skip reinstallation. An asterisk marks the active environment. A missing conda command in an ordinary terminal can mean that shell is not initialized, rather than that conda is absent.

如果都成功，就跳过重新安装。环境列表中的星号表示当前激活环境。普通终端提示找不到 conda，可能只是该终端尚未初始化 (shell initialization)，不能直接认定未安装。

Screenshot placeholder 07-1 — Anaconda Prompt showing conda --version and conda env list. Suggested file: `docs/images/07-conda-available.png`

插图占位 (Screenshot placeholder) 07-1 — Anaconda Prompt 中的 conda --version 与 conda env list 输出。建议文件： `docs/images/07-conda-available.png`

If no conda installation exists, use the [official Windows installation guide](#)^[1] and a matching installer from [Anaconda's download page](#)^[2]. Miniconda provides a smaller starting installation; Anaconda Distribution includes more preinstalled packages. For a personal machine, use a per-user installation where appropriate. Finish the installer, open a fresh Anaconda Prompt, and repeat the two checks above. Keep environments outside your Git repository.

如果确实没有 conda，按官方 Windows 安装指南^[1]，从 Anaconda 下载页^[2]选择适合系统的安装程序。Miniconda 初始安装较小；Anaconda Distribution 预装的软件包更多。个人电脑适合时可采用当前用户安装（per-user installation）。安装结束后重新打开 Anaconda Prompt，再执行上面的检查。不要把整个环境装进 Git 仓库。

2. Create a project environment / 2. 创建项目环境 (Create environment)

The example chooses Python 3.11 for an isolated teaching environment. For a real project, use the version required by its dependencies or its existing environment file. First make sure handbook-demo is not already listed; if it exists, inspect it or choose a different name.

教学示例使用 Python 3.11；真实项目应按照依赖要求或已有环境文件选择版本。先检查列表中没有 handbook-demo；如果已存在，先检查它或另选名称。

Run each command after the previous one succeeds, and review conda's proposed package changes when prompted:

逐条执行，上一条成功后再继续；conda 提示确认时，查看拟安装的软件包：

```
conda create -n handbook-demo python=3.11 numpy matplotlib pip
conda activate handbook-demo
python --version
python -c "import sys; print(sys.executable)"
python -c "import numpy, matplotlib; print(numpy.__version__, matplotlib.__version__)"
```

Check that `sys.executable` points into the new environment. A successful import verifies that both packages are available to that interpreter. Do not rely only on the prompt prefix. See [conda environment management](#)^[3].

`sys.executable` 应指向新环境的 Python。导入 (import) 成功说明该解释器能够使用 NumPy 和 Matplotlib；不要仅凭命令行前缀判断。参见 [conda 环境管理文档](#)^[3]。

Screenshot placeholder 07-2 — Activated handbook-demo environment and interpreter/version verification.

Suggested file: docs/images/07-conda-active.png

插图占位 (Screenshot placeholder) 07-2 — handbook-demo 环境已激活，并显示解释器路径与版本检查结果。

建议文件：docs/images/07-conda-active.png

`conda activate handbook-demo` selects an existing environment; it does not install anything. `conda deactivate` leaves it. If activation fails in PowerShell, run `conda init powershell` from Anaconda Prompt, then close and reopen PowerShell. This initializes that shell's startup configuration. If local policy prevents startup scripts, use Anaconda Prompt rather than changing system-wide security settings as a first response.

`conda activate handbook-demo` 是激活已有环境，不会安装软件；`conda deactivate` 退出当前环境。如果 PowerShell 无法激活，可在 Anaconda Prompt 执行 `conda init powershell`，然后关闭并重新打开 PowerShell。该命令会配置终端启动文件。若本机策略禁止启动脚本，可先使用 Anaconda Prompt，不要一开始就更改系统范围的安全设置。

3. Select the interpreter in VS Code / 3. 在 VS Code 选择解释器 (Select interpreter)

Install Microsoft's Python extension if it is not already available. Open the project, press **Ctrl+Shift+P**, run **Python: Select Interpreter**, and select handbook-demo. If necessary, enter the interpreter path printed by `sys.executable`. Open a new terminal and repeat the interpreter/import checks. Existing terminals may still have the previous environment active. See [VS Code's environment guide](#)^[4].

如果尚未安装，先安装 Microsoft 的 Python 扩展 (extension)。打开项目，按 **Ctrl+Shift+P**，选择 **Python: Select Interpreter**，再选择 handbook-demo。如果找不到，可输入前面 `sys.executable` 显示的路径。新建终端后，再检查解释器及导入结果；已有终端可能仍使用旧环境。参见 [VS Code 环境指南](#)^[4]。

Screenshot placeholder 07-3 — VS Code interpreter picker with handbook-demo selected. Suggested file:

`docs/images/07-vscode-interpreter.png`

插图占位 (Screenshot placeholder) 07-3 — VS Code 解释器选择列表，突出显示 handbook-demo 已选中。建议文件：`docs/images/07-vscode-interpreter.png`

For a notebook, its kernel is a separate selection: choose the intended environment through **Select Kernel**. If it needs an IPython kernel, install `ipykernel` in that environment and select it again:

Notebook 的内核 (kernel) 需要单独选择：通过 **Select Kernel** 选择目标环境。如果该环境缺少 IPython 内核，可执行后重新选择：

```
conda install -n handbook-demo ipykernel
```

4. Save a reproducible starting specification / 4. 保存可复现的环境说明 (Environment specification)

Create a UTF-8 `environment.yml` in your project, for example:

在项目中创建 UTF-8 编码的 `environment.yml`，例如：

```
name: handbook-demo
channels:
  - defaults
dependencies:
  - python=3.11
  - numpy
  - matplotlib
  - pip
```

To reproduce this specification in a fresh environment, use `conda env create -f environment.yml`. If the name already exists, use a new one, for example `conda env create -n handbook-demo-copy -f environment.yml`. This recipe intentionally leaves most package versions open; it is not an exact lock file. Record the resolved versions when comparing research results.

在新环境中重建时执行 `conda env create -f environment.yml`。若同名环境已存在，改用新名称，例如 `conda env create -n handbook-demo-copy -f environment.yml`。这个配方没有固定大部分包版本，并非精确锁定文件 (lock file)；比较科研结果时，应同时记录实际解析得到的版本 (resolved versions)。

To export the current environment's explicitly requested packages in PowerShell:

在 PowerShell 导出该环境中明确请求安装的软件包：

```
conda env export -n handbook-demo --from-history | Set-Content -Encoding utf8 environment.yml
```

This overwrites that file; use a different filename if you want to retain the hand-written recipe. Inspect the export before sharing: remove any machine-specific `prefix: path`. The history export is useful for portability but does not capture every resolved dependency or every package installed with pip. Channel choice, operating system, and package availability also affect reconstruction.

命令会覆盖同名文件；若要保留手写配方，请改用其他文件名。分享前检查导出内容，删除与本机有关的 `prefix: 路径`。历史导出（history export）有利于跨机器使用，但不会完整记录所有间接依赖，也不能保证包含通过 pip 安装的所有包。软件源（channel）、操作系统及包的可用性也会影响重建结果。

If pip is needed, install conda packages first and then use `python -m pip install PACKAGE_NAME` inside the activated environment, replacing `PACKAGE_NAME` with the actual package. Record these additional dependencies; prefer rebuilding from a complete specification when changing a mixed conda/pip environment.

如需 pip，先安装 conda 包，再在激活的环境中使用 `python -m pip install PACKAGE_NAME`，将 `PACKAGE_NAME` 替换为实际包名，并记录新增依赖。后续调整 conda/pip 混合环境时，优先根据完整说明重新构建。

5. Troubleshooting and completion check / 5. 排查与完成标准 (Troubleshooting and verification)

Symptom 现象 (symptom)	First check 首先检查 (first check)
conda not recognized 找不到 conda	Use Anaconda Prompt; check shell initialization 使用 Anaconda Prompt，检查终端初始化
ModuleNotFoundError	Print <code>sys.executable</code> ; install in that environment 查看 <code>sys.executable</code> ，在对应环境安装包
VS Code runs another Python VS Code 使用了另一个 Python	Select the interpreter, then open a new terminal 重新选择解释器，再新建终端
Notebook uses another environment Notebook 环境不一致	Check its kernel selection 检查内核选择 (kernel selection)
Dependency solving fails 依赖解析失败 (solving fails)	Check Python/package compatibility and project requirements 核对 Python、包版本与项目要求

You are ready when the environment is listed, its interpreter imports the required packages, VS Code uses that interpreter, and the project has a reviewed environment specification.

环境已出现在列表中、其解释器能够导入所需包、VS Code 使用同一解释器、项目保存了已检查的环境说明，就完成了基本配置。

6. Replace the screenshot placeholders / 6. 后续替换插图 (Screenshot placeholders)

Save future screenshots under the three suggested names in `docs/images/`. Replace each visible block with the image line in its adjacent HTML comment, removing the comment markers. The bilingual guides share those image files; no nonexistent images are linked yet. Crop personal paths if sharing screenshots publicly.

将截图保存到 docs/images/ 下的三个建议文件名，用相邻 HTML 注释里的图片语句替换可见占位区块，并删除注释标记。双语指南共用图片；当前未链接不存在的图片。公开截图时可裁去个人路径。

Regenerate with `python tools/export_print.py docs/07-main-conda-environment.md`.

重新运行 `python tools/export_print.py docs/07-main-conda-environment.zh-CN.md` 即可更新打印版。

参考资料 (References)

1. <https://docs.conda.io/projects/conda/en/latest/user-guide/install/windows.html>
2. <https://www.anaconda.com/download>
3. <https://docs.conda.io/projects/conda/en/stable/user-guide/tasks/manage-environments.html>
4. <https://code.visualstudio.com/docs/python/environments>